

# An Online Self-Correcting Calibration Architecture for Multi-Camera Traffic Localization Infrastructure

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**Abstract**—Most vision-based sensing and localization infrastructure today employ conventional area scanning cameras due to the high information density and cost efficiency offered by them. While the information-rich two-dimensional images provided by such sensors make it easier to detect and classify traffic objects with the help of deep neural networks, their accurate localization in the three-dimensional real world also calls for a reliable calibration methodology, that maintains accuracy not just during installation, but also under continuous operation over time. In this paper, we propose a camera calibration architecture that extracts and uses corresponding targets from high definition maps, augment it with an efficient stabilization mechanism in order to compensate for the errors arising out of fast transient vibrations and slow orientational drifts. Finally, we evaluate its performance on a real-world test site.

## I. INTRODUCTION

By creating a digital traffic twin that accurately reflects the live traffic situation, smart infrastructures can provide situational awareness to drivers and autonomous cars, improving their understanding of the current traffic environment. Vision-based systems have become increasingly popular for implementing smart traffic surveillance, largely because cameras offer a high information content combined with a continuous cost reduction. However, a key challenge lies in the reliable and accurate calibration of the roadside sensors. It is crucial for reconstructing and fusing the 3D scene from the camera images, and for providing precise 3D world coordinates of the traffic objects which is a requirement for most traffic applications.

A common method for retrieving the camera pose of a vision-based system is the minimization of the reprojection error using image features and selected calibration targets or patterns [1]. Furthermore, in the context of calibrating vehicle-mounted sensors, Knorr *et al.* [2] use ground plane observations to estimate the motion and pose parameters of a moving multi-camera system through recursive filtering with dedicated motion models. Many calibration methods require the motion of the sensors themselves for the recovery of the camera pose [3], [4]. While those approaches are suitable for the calibration of vehicle sensors they are not adequate for the calibration of the nearly-stationary sensors mounted to road infrastructure.

Various different methodologies have been specifically developed for the calibration of roadside sensors. A large portion of them are not suitable for localization applications because they use vanishing points [5], [6] or vehicle trajectories [7] instead of geo-referenced calibration targets. In the recent past, very promising approaches in terms of accuracy and robustness have been presented solving the geo-referenced calibration of infrastructure sensors [8], [9], [10], [11], [12]. They use vehicles as calibration targets and retrieve the camera pose based on the recorded vehicle localization as they move through the sensors' fields of view (FOVs). These calibration schemes offer automatic calibration solutions with great potential for the application in infrastructure systems. However, they do not address the long- or short-term stability of the calibration during operation.

Maintaining the accuracy of the calibration during operation while subjected to real-life disturbances such as system vibrations or insufficient mounting rigidity remains a challenge. To address this issue, we propose a self-correcting calibration architecture for infrastructure-based sensor systems. First, the cameras are calibrated offline against geo-referenced targets extracted from a high definition (HD) map to establish an accurate baseline calibration. Second, we employ an online correction approach based on image stabilization methods that complement the baseline calibration, addressing errors arising during operation. In contrast to general image stabilization applications that carry out costly image warping [13], [14], [15], we incorporate the stabilizing transformations directly into the 2D to 3D coordinate conversion applied to the output of the object detectors. In this manner, the stabilization can be positioned in parallel to the detectors, avoiding a serial processing step. Our proposed architecture offers great flexibility, as the baseline calibration with the HD map can be exchanged or extended by other methods that provide geo-referenced calibration and it can further be paired with any real-time capable image stabilization algorithm for the online correction.

The paper is organized as follows: In Section II, we start with a description of the proposed system architecture, in particular with regard to the calibration methodology. Thereafter, in Section III, we evaluate its localization accuracy using a real-world infrastructure system (Fig. 1) established on the highway between Munich and Nuremberg during the research projects *Providentia* and *Providentia++* [16], [17], [18]. Finally, we summarize our results in Section IV.

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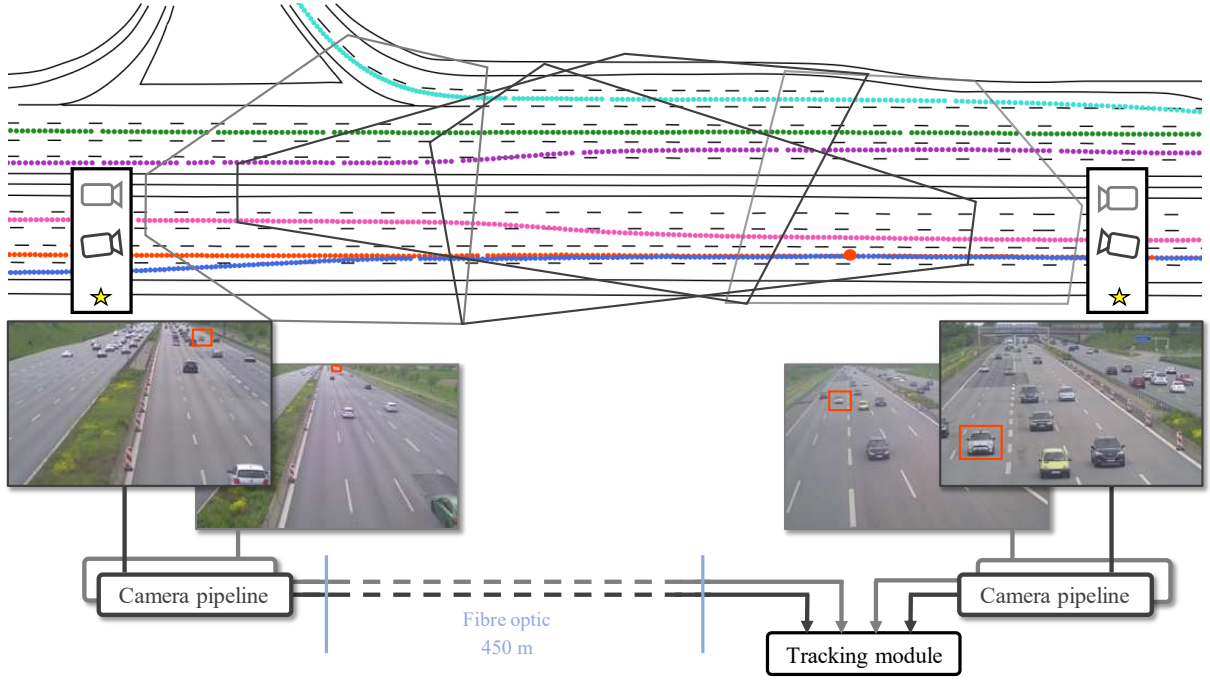


Fig. 1: Overview of the multi-camera infrastructure system used as the experimental setup in this work. The outer gantry piers with a distance of approximately 450 m indicated by stars are located at N48°14'29.796" E11°38'21.083" (left) and N48°14'16.170" E11°38'14.251" (right). The FOVs of the cameras are outlined and exemplary concurrent images are shown. The ground truth tracks are displayed as dotted paths. The vehicle with the GNSS device mounted to its top is highlighted in the example images and its approximate position is marked. Additionally, the basic data processing chain is visualized.

## II. SYSTEM ARCHITECTURE

In this section, we describe the architecture of our proposed traffic localization system. The system incorporates multiple cameras that monitor the region of interest (ROI) from different viewpoints (see Fig. 1). All the data from this multi-view setup is combined into one consistent representation of the real-time traffic situation. In order to minimize the data transfer between computing nodes and optimize processing efficiency, the raw images obtained from the cameras are processed locally using parallelized camera pipelines. These pipelines condense the raw images into object lists which are in turn transmitted to the tracking module that runs on a designated node within the traffic surveillance system. Following the fundamental principle of tracking-by-detection, the module tracks the location of all traffic participants in a common world frame within the ROI. By processing images locally and only transferring the necessary object lists, the proposed approach effectively reduces the amount of data needed to be transmitted between nodes, thereby minimizing the communication overhead.

### A. Camera Processing Pipeline

As the first step in the camera processing pipeline, a detection and classification network is employed for vehicle detection within the RGB-encoded camera images. Because of its good performance in terms of speed and accuracy, we choose the pretrained YOLOv7 detector for this task [19]. The network processes each image to identify and classify vehicles based on their visual characteristics. The detections

are represented using axis-aligned bounding boxes, which delineate the object's position within the image.

In a subsequent postprocessing step, the image detections are transformed to locations in a Cartesian world frame. By performing this conversion, objects can be tracked effectively in world coordinates and data from sensors observing different parts of the ROI can be seamlessly fused within the common frame. The reconstruction from the 2D image to the 3D scene utilizes the projective camera model [20]. The calibration matrices are derived during the calibration phase, which is described in the following section. Additionally, to estimate the center coordinates of each detected object, we employ an algorithm for fitting 3D cuboids to 2D bounding boxes. This approach, published in [21], enables us to transmit the center coordinate on the road-plane of the fitted cuboid as the object's location and, in addition, provides us with estimates of its extents.

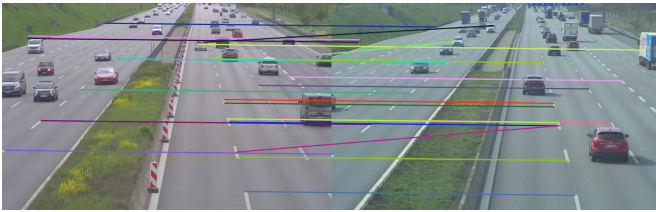
### B. Baseline Camera Calibration

The projective camera model is an established tool in computer vision for reconstructing the 3D world from 2D images [20]. The relationship between 3D world coordinates  $[x, y, z]^T \in \mathbb{R}^3$  and their corresponding 2D image coordinates  $[u, v]^T \in \mathbb{R}^2$  is defined by the projection matrix  $P \in \mathbb{R}^{3 \times 4}$ . Using homogeneous coordinates, the relation is given by

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = P \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}.$$



Fig. 2: Points from the HD map (left) are mapped to points in the image (right). These correspondences are used to compute the optimal camera calibration. With this result, the digital lane markings are visualized in the calibrated image (center).



(a) Detected and matched features between the input image (left) and the calibrated reference image (right).



(b) Frame misalignment caused by a camera pose shift. (c) Corrected transformation ensures perfect alignment.

Fig. 3: Online calibration correction based on image stabilization methods.

Let  $R \in \mathbb{R}^{3 \times 3}$  be the extrinsic camera rotation and  $t \in \mathbb{R}^3$  the corresponding translation. The projection matrix  $P$  is constructed from the intrinsic camera matrix  $K \in \mathbb{R}^{3 \times 3}$  and the extrinsic transformation  $[R | t] \in \mathbb{R}^{3 \times 4}$ , as follows

$$P = K[R | t].$$

A common method for recovering the camera pose and the intrinsic parameters is to minimize the reprojection error with respect to a set of correspondences of world points and their respective pixel coordinates [20], [1], [22].

In our infrastructure system, the sensors are calibrated against a dedicated HD map. This HD map contains an accurate representation of the environment in the Universal Transverse Mercator (UTM) system. Thus, we formulate our global reference frame in relation to the UTM system to allow for an easy transformation between the frames. The map includes, among other things, the precise coordinates of the road markings. The dashed white line markings, which are very common lane indicators on German roads, are available

with their start and end coordinate. Using classical image processing algorithms, the corresponding lane markings are extracted from the image and mapped to their counterparts in the HD map. The resulting set of correspondences of world points and their respective image points are the basis for computing the optimal camera calibration using the reprojection error minimization method (see Fig. 2).

This process is carried out once offline prior to the deployment of the system. It provides us with the camera calibration information which is necessary for the transformation of the image detections in the postprocessing step. We also store the calibrated reference image which plays a pivotal role in the online correction module.

### C. Online Calibration Correction

To address errors in the camera calibration that may arise over time or due to system vibrations, we employ an online correction approach based on image stabilization methods. This technique corrects inconsistencies by aligning the input images to a reference image. Here, the calibrated image of the previously described camera calibration process serves as the reference image for the stabilization. We detect and match SURF features [23] in both the input and the calibrated reference image, as displayed in Fig. 3a. With the set of feature matches, we estimate the homographic transformation  $H \in \mathbb{R}^{3 \times 3}$  that aligns the feature locations of the input image to their correspondences in the reference image. This is accomplished by following a well-established procedure that again minimizes the reprojection error and utilizes the RANSAC method [20], [24].

Typical video stabilization applications aim to compensate the detected motion directly in the consecutive images. This is achieved by warping the images using the computed homographic transformation. However, placing image stabilization and warping before the detector would add an unnecessary computationally intensive serial processing step, increasing latency. Instead, the online correction module operates in parallel to the detector, allowing both to process the captured image simultaneously. See the data flow diagram in Fig. 4 for a visualization of this concept. Subsequently,

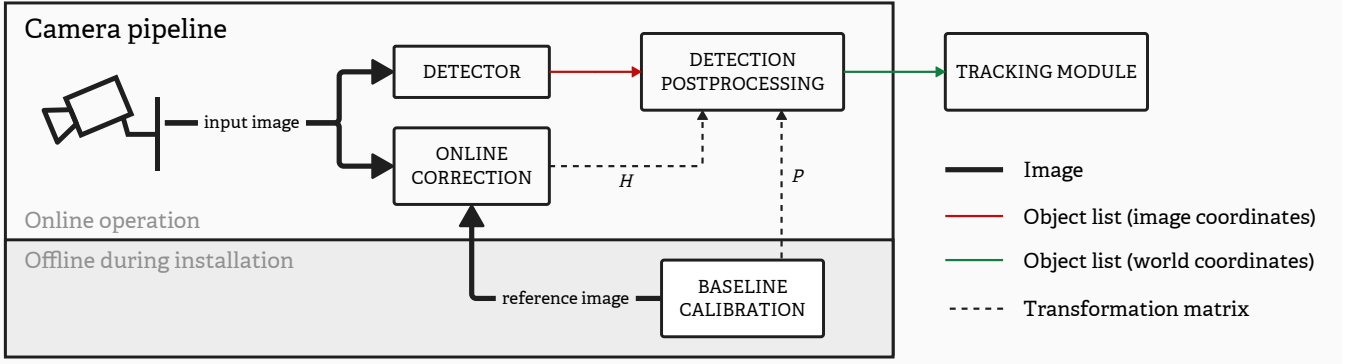


Fig. 4: Proposed system architecture and data flows for the camera processing chain.

the stabilization matrix  $H$  is forwarded to the detection post-processing module and incorporated in the transformation of the 2D detections generated from the respective image. The projection matrix is updated as

$$P' = HP = HK[R \mid t].$$

Fig. 3b demonstrates a shift in the camera pose that causes misalignment between the road markings in the input image and the HD map visualization based on the baseline calibration. The disparity between the world and the image frame arises due to the camera's new position relative to the environment. By incorporating the stabilizing homography computed specifically for this image, the HD map aligns perfectly with the image in Fig. 3c. Using the corrected transformation ensures accurate object localization.

#### D. Tracking module

The postprocessed detection data streams of the cameras are all processed and fused in the order of their recording by a single tracking module allocated to the ROI. The multi-object tracker continuously estimates the state of the current traffic situation, while factoring in measurement errors and uncertainties. We employ a Labeled Multi-Bernoulli (LMB) tracker [25] and model all spatial and kinematic parameters to be normally distributed, resulting in a Gaussian mixture for the spatial distribution of the components within the multi-object state. This allows for efficient calculations in the steps of the tracker using the equations of the Kalman filter. The kinematic state  $\hat{x} = [x, y, v_x, v_y]^T \in \mathbb{R}^4$  consists of the position  $x, y$  and the Cartesian velocity  $v_x, v_y$ . It is assumed to follow a constant velocity motion model [26]. In our previously published work [21], we present our tracking module in more detail and provide further information on the parameter choices.

### III. EXPERIMENTS

In this section, we present experimental evaluations to assess the effectiveness of our online calibration correction approach. As a proof of concept, we conduct experiments designed to investigate the compensation of camera pose drifts that develop over time and of high-frequency disturbances

caused by vibrations in the system. We compare three distinct system configurations:

- 1) The first configuration C1 employs the baseline calibration without online correction.
- 2) Configuration C2 computes the correction only for the first image in the dataset and applies it as a constant correction to subsequent images. This way, the ability to correct long-term degradation of the calibration is assessed.
- 3) In the third configuration C3, we dynamically compute the correction for every image in the dataset to evaluate the benefits in the presence of image vibrations.

#### A. Dataset

The infrastructure system as visualized in Fig. 1 serves as the experimental setup. The captured images from four cameras observing the highway from different viewpoints are recorded at approximately 20 Hz to generate the dataset for the experiments. The baseline calibration of these four cameras was computed over one year prior to the date of the data recording. To evaluate the localization precision for the three system configurations, we employ a vehicle equipped with a high-precision Global navigation satellite system (GNSS) device<sup>1</sup> and record six ground truth paths through the ROI. The speed of the recorded vehicle was on average  $84.3 \text{ km h}^{-1}$ . Fig. 1 illustrates these ground truth paths where each point represents a recorded GNSS position transformed to our global reference frame as well as the four cameras' FOVs within the system environment.

#### B. Results

The recorded images are fed into and processed by our system, with the alteration of the three online correction setups as described. We extract the estimated vehicle location from the output of our tracking module at the same time stamps as the synchronized ground truth positions. By calculating the Euclidean distance between these matched ground truth and estimated positions for each time step in

<sup>1</sup>Emlid Reach RS2 using Real-Time Kinematic (RTK) correction provided by the real-time positioning service SAPOS resulting in centimeter-level precision.

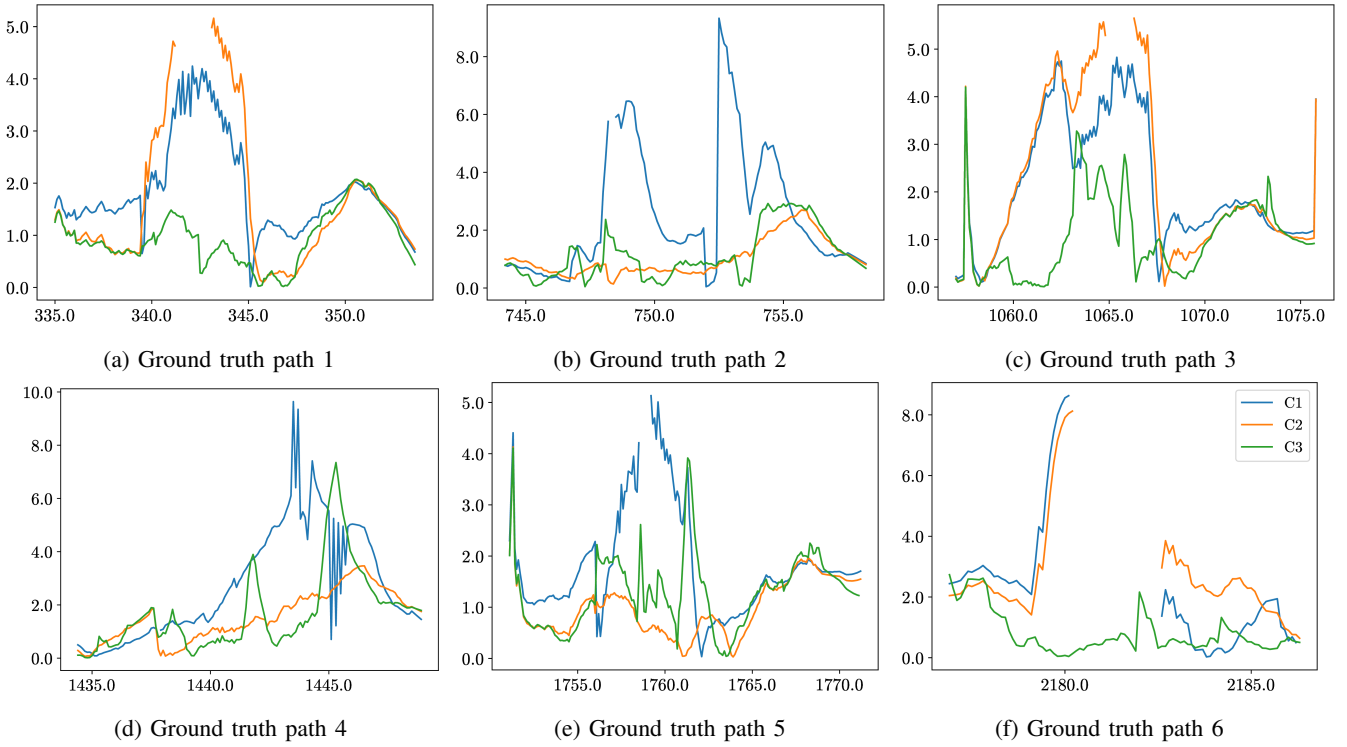


Fig. 5: Position error [m] over time [s] since the start of the recording for the six ground truth paths. Interruptions in the graph indicate the absence of a matching track output from the system.

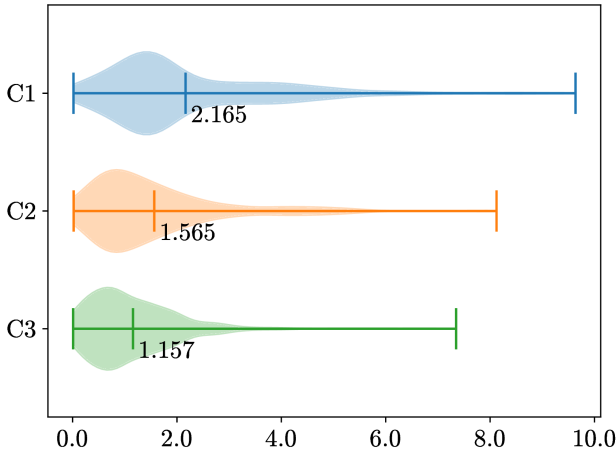


Fig. 6: Distribution of the position error [m] of all paths for the three system configurations. The minimum, mean and maximum values are indicated with horizontal lines. The value of the mean position error is specified.

the same reference frame, we obtain quantifiable position error metrics to evaluate the effectiveness of our approach.

In Fig. 5, we visualize the evaluation results of the experiments. The figure shows the time series of the position error comparing the three system configurations broken down into the six ground truth paths. We can observe that the green graph representing the dynamic correction setup (C3) is almost always below the baseline (C1) shown in blue.

The improvement is evident when comparing the position errors in all the plots. This demonstrates the advantages of dynamically adapting to the changing camera poses and continuously minimizing the errors. In the case of applying a constant correction (C2), the results are not as clear-cut. While some plots (Figs. 5b, 5d and 5e) indicate a substantial improvement towards the baseline, the others show very large remaining errors (Figs. 5a, 5c and 5f). The errors at times also exceed the baseline while additional track losses are also observed. The accuracy of the correction depends heavily on the image selected for computing the constant correction and how well it aligns with subsequent images.

Further, we analyze the distribution of the position error values displayed in Fig. 6. The first thing to note is that both correction setups (C2 & C3) in general reduce the error both in terms of maximum and mean values compared to the baseline (C1). Even though an improvement is not guaranteed for all time steps in case of the constant correction method as discussed earlier, it still delivers better results on average in this experiment. The benefit of the dynamic stabilization configuration is again clearly apparent as the mean position error is significantly reduced from 2.165 m as the baseline without any correction to 1.157 m. These findings validate the efficacy of our online correction approach, particularly when dealing with environmental changes and vibrations that can affect the camera calibration.

## IV. CONCLUSION

In this paper, we proposed an architecture for vision-based infrastructure systems and in particular we presented an approach for calibrating its cameras. On top of the baseline calibration which is carried out offline, an online correction module compensates calibration drifts and system vibrations. As a proof of concept, we conducted experiments involving a vehicle equipped with a high-precision GNSS device for recording ground truth paths within the system. We analyzed the localization accuracy of our proposed system with three distinct system configurations. Our experimental results demonstrated significant improvements in the localization precision for both configurations involving online correction compared to the baseline without online correction. The most notable enhancement was observed with the dynamic online correction, which effectively countered the errors for the individual images. These findings validate the potential and the effectiveness of our presented architecture in delivering accurate and robust localization for infrastructure systems exposed to real-world disturbances.

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